

SustainAlign

An AI-Powered Platform for Corporate Social Responsibility & ESG Alignment in India

1. Problem Statement

India is the **only country in the world with a legally mandated CSR law** (Section 135, Companies Act 2013), requiring eligible companies to spend 2% of average net profit on Corporate Social Responsibility. The annual CSR spend stands at **Rs. 34,909 crore (FY24)**, yet over **Rs. 7,000+ crore goes unspent every year** due to three systemic failures:

- **Discovery Gap:** Corporates cannot locate credible, verified NGOs aligned with their budget, sector, and SDG priorities.
- **Compliance Chaos:** Fragmented reporting across MCA, SEBI, GRI, SASB, and UN SDGs creates audit risk and administrative overhead.
- **Impact Verification:** NGOs lack standardized tools to prove outcomes; corporates cannot quantify social return on investment.

2. Proposed Solution

SustainAlign is a full-stack, multi-tenant SaaS platform that connects **Corporates, NGOs, and Regulators** through intelligent project discovery, AI-powered matching, compliance automation, and real-time impact tracking. It serves as the **"LinkedIn + Bloomberg Terminal" for CSR and ESG in India**.

3. Key Features

Module	Capability
AI Copilot	Natural-language CSR assistant powered by Ollama (Llama 3.1); answers compliance queries, generates reports, recommends projects.
RAG-Based NGO Discovery	Vector embeddings (nomic-embed-text) with cosine similarity search for context-aware NGO matching beyond keyword search.
Explainable AI Matching	Budget-to-SDG alignment scoring with transparent rationale; generates "Top 3" recommendations with justification.
Compliance Engine	Automated MCA, SEBI BRSR, GRI, SASB, and UN SDG reporting with audit-proof document generation (PDF/DOCX/PPTX).
Impact Tracking	Live KPI dashboards, beneficiary logging, geo-tagged updates, and milestone tracking for ongoing CSR projects.
Volunteer Management	Event creation, QR-based check-in, attendance tracking, and automated certificate generation.

4. Technical Architecture

SustainAlign follows a **monorepo architecture** (npm workspaces) with a React 19 frontend and Express 5 backend (ESM modules). The system is **multi-tenant** with row-level data isolation via tenant IDs across 50+ database tables. Role-Based Access Control enforces 44+ granular permissions across corporate, NGO, and platform admin portals.

- **ORM & Database:** Drizzle ORM with SQLite (development) and PostgreSQL 16 (production) with PgBouncer connection pooling.
- **Authentication:** Argon2 password hashing, JWT (jose) with rotating refresh tokens, MFA support, and role-based invitation workflows.
- **AI/ML Pipeline:** Ollama for LLM inference (Llama 3.1) and vector embeddings (nomic-embed-text). Graceful offline degradation with structured fallback scoring.
- **Infrastructure:** Docker + Docker Compose, Nginx reverse proxy with WAF rules, Terraform IaC (DigitalOcean), Ansible provisioning, GitHub Actions CI/CD.
- **Observability:** Prometheus + Grafana metrics, Loki + Promtail log aggregation, Sentry error tracking, Pino structured logging.

- **Security:** Helmet.js (CSP, HSTS), tiered rate limiting, CORS, Zod validation on all endpoints, Gitleaks + GitGuardian secret scanning.

5. Technology Stack

Layer	Technologies
Frontend	React 19.2, Vite 8, Tailwind CSS 4.3, React Router 7, Recharts, React Hook Form + Zod
Backend	Express 5, Drizzle ORM 0.45, Zod 3.24, jose JWT, argon2, PDFKit, docx, pptxgenjs
AI / ML	Ollama (Llama 3.1:1b + nomic-embed-text), cosine similarity vector search, RAG pipeline
Database	SQLite (dev), PostgreSQL 16 (prod), PgBouncer connection pooling
Infrastructure	Docker, Nginx, Terraform, Ansible, GitHub Actions, Prometheus, Grafana, Loki, Sentry
Testing	Vitest, Supertest, Testing Library, CI: lint + build + test (SQLite + PG)

6. Innovation & Impact

What sets SustainAlign apart: It is not a concept — it is a **working, production-hardened prototype** that has completed 7 production-readiness phases spanning CI/CD, database migration, API documentation, and security hardening. The AI layer operates offline with graceful degradation, ensuring reliability in low-connectivity environments common in rural NGO operations.

- **Market:** Rs. 34,909 crore annual CSR spend (FY24) with 28,000+ reporting companies.
- **Pricing:** SaaS tiers from Rs. 49,999/month (Starter) to custom Enterprise plans.
- **Target Segments:** PSUs, BFSI, IT corporates, and mid-cap companies facing CSR shortfalls.
- **Compliance Coverage:** MCA (India), SEBI BRSR, GRI Standards, SASB, UN SDGs — covering 90%+ of mandatory CSR reporting requirements.
- **Licensing:** MIT Licensed — open for adoption, customization, and community contribution.

7. Conclusion

SustainAlign addresses a **legally mandated, billion-dollar market gap** with a technically sophisticated, AI-driven solution. The platform demonstrates production-grade engineering across the full stack — from multi-tenant data isolation and 44+ RBAC permissions to RAG-based NGO discovery and automated compliance reporting. With a working prototype, comprehensive test suite, and containerized deployment pipeline, SustainAlign is positioned for immediate pilot deployment with corporate CSR departments and NGO partners.